

Answer any **four** questions. Each question has a total of 20 marks.

- 1** Hydrolysis involves the cleavage of chemical bonds on reaction with water. It is often used in the synthesis of important chemical intermediates.

- (a)** The rate of the hydrolysis of bromoalkanes with aqueous potassium hydroxide depends upon the chemical structure of the bromoalkane.

- (i)** Explain what is meant by the **rate** of a chemical reaction.

The table below shows the results of a series of hydrolysis experiments that was carried out using different concentrations of the bromoalkane, $\text{C}_4\text{H}_9\text{Br}$, and aqueous KOH.

<i>experiment</i>	<i>initial</i> $[\text{C}_4\text{H}_9\text{Br}] / \text{mol dm}^{-3}$	<i>initial</i> $[\text{KOH}] / \text{mol dm}^{-3}$	<i>initial rate</i> $\text{mol dm}^{-3} \text{s}^{-1}$
I	0.10	0.10	0.024
II	0.15	0.10	0.036
III	0.20	0.20	0.048

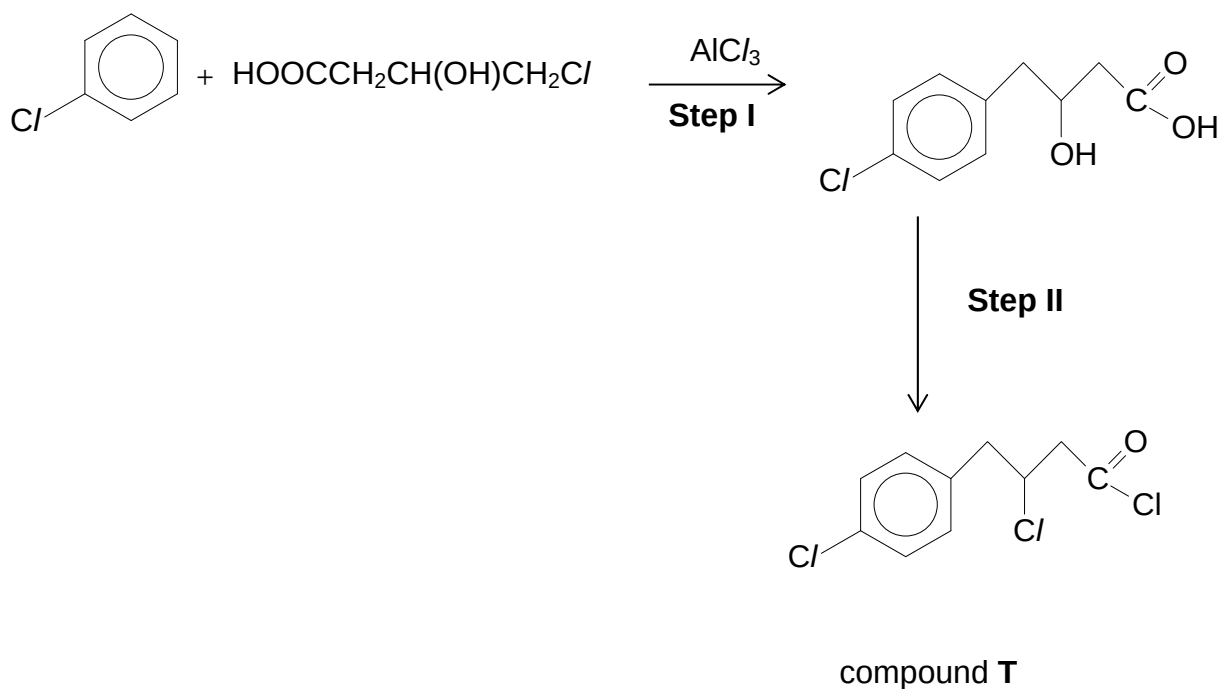
- (ii)** Use the given data to deduce the order of reaction with respect to each of the two reactants in the hydrolysis reaction, showing how you arrive at your answers.
- (iii)** Hence, write the rate equation for the reaction and state the units of the rate constant.
- (iv)** Given that the bromoalkane is achiral, use your answer in **(a)(iii)** to suggest a possible structural formula for the bromoalkane.

Hence, propose a mechanism for the hydrolysis of the bromoalkane in aqueous potassium hydroxide that is consistent with the rate equation.

- (v)** How would you expect the rate of hydrolysis to change if the bromoalkane was replaced by an iodoalkane of similar structure? Explain your answer with the use of suitable values from the *Data Booklet*.
- (vi)** In a separate experiment, the bromoalkane from **(a)(iv)** is treated with alcoholic KOH. State the type of reaction that occurs and give the structural formula of the organic product formed.

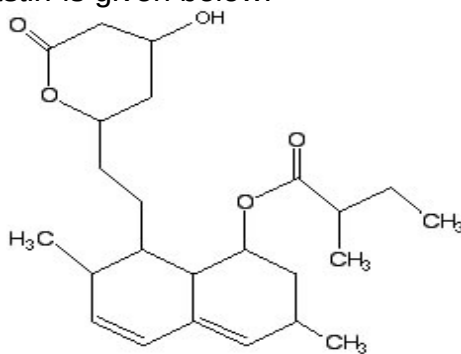
[12]

(b) Consider the reaction scheme below.



- State the type of reaction in **Step I**.
- State the reagent(s) and condition(s) required for **Step II**.
- Explain the difference in the reactivities of the 3 chlorine atoms in compound T. [5]

(c) The structure of *lovastatin* is given below.

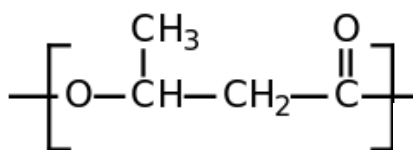


lovastatin

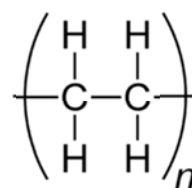
Draw the structure of the products formed when *lovastatin* reacts with hot, aqueous sodium hydroxide.

[2]

- (d) The structures below show small sections of the polymers – *polyhydroxybutyrate* and *polyethene*. Both polymers are used in the manufacture of films for packaging.



Polyhydroxybutyrate



polyethene

(n > 500)

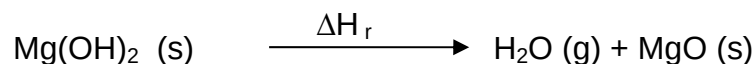
Unlike polyethene, *polyhydroxybutyrate* degrades within 4 weeks in landfills. Suggest an explanation for this.

[1]

[Total: 20]

2 This question is about magnesium and its compounds.

- (a) Magnesium hydroxide can be used as a fire-retardant as it is able to decompose upon heating at 332°C according to the following equation:



standard enthalpy change of atomisation of magnesium	+ 147 kJ mol ⁻¹
1 st electron affinity of oxygen	– 142 kJ mol ⁻¹
2 nd electron affinity of oxygen	+ 844 kJ mol ⁻¹
lattice energy of magnesium oxide	– 3920 kJ mol ⁻¹
standard enthalpy change of formation of water	– 289 kJ mol ⁻¹
standard enthalpy change of vaporisation of water	+ 22.3 kJ mol ⁻¹
standard enthalpy change of formation of magnesium hydroxide	– 927 kJ mol ⁻¹

- (i) Explain the difference between the first and second ionisation energies of magnesium.
- (ii) Suggest why the second electron affinity of oxygen is endothermic.
- (iii) Using information from the table above, together with relevant data from the *Data Booklet*, draw a fully labelled energy level diagram and calculate the standard enthalpy change of formation of solid magnesium oxide.
- (iv) Using your answer in (a)(iii) and relevant values provided in the table above, calculate ΔH_r , the enthalpy change for the thermal decomposition of solid magnesium hydroxide. State an assumption made in your calculations.
- (v) By considering your answer in (a)(iv), suggest how magnesium hydroxide functions as a fire-retardant.

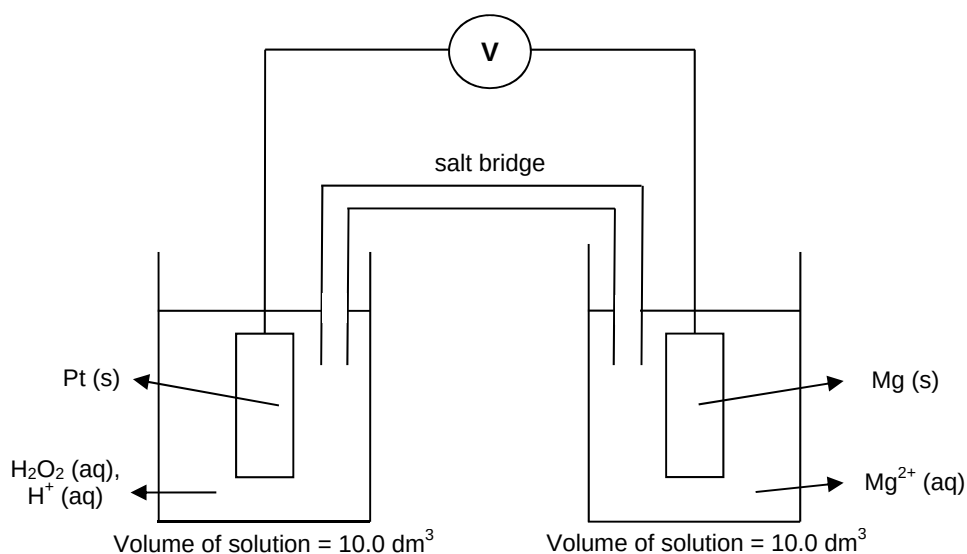
- (vi) Barium hydroxide undergoes a thermal decomposition reaction similar to that of magnesium hydroxide.

By quoting suitable data from the Data Booklet, deduce whether barium hydroxide will decompose at a higher or lower temperature than magnesium hydroxide. Explain your answer clearly.

[13]

- (b) A magnesium-hydrogen peroxide fuel cell has been developed to power undersea vehicles.

The following diagram is a simplified representation of the cell.

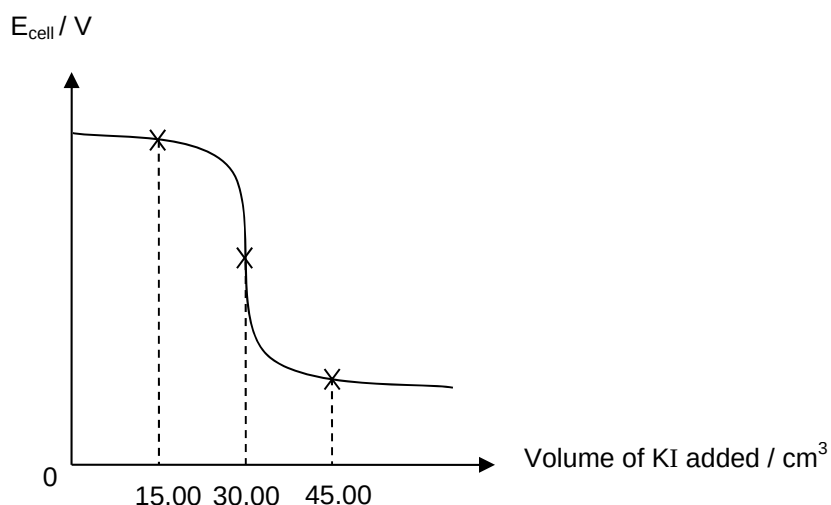


The energy derived from this fuel cell depends on the concentration of H_2O_2 used.

A 25.0 cm^3 sample of the aqueous hydrogen peroxide solution was taken from the cell and its concentration was estimated by titration with 1.50 mol dm^{-3} aqueous potassium iodide.

The E_{cell} was measured against a standard hydrogen electrode as aqueous potassium iodide was added.

The following graph was obtained:



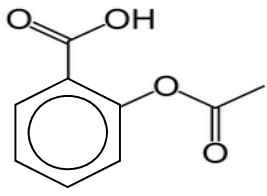
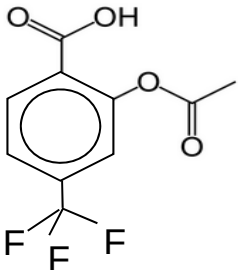
- (i) Write a balanced equation for the reaction between potassium iodide and acidified H_2O_2 .
- (ii) Calculate the concentration of hydrogen peroxide in the 25.0 cm^3 sample.
- (iii) Determine the minimum mass of Mg metal required at the anode in the fuel cell to fully utilise the hydrogen peroxide present in the cell shown on page 5.
- (iv) Suggest why calcium is not a suitable replacement for magnesium in this fuel cell set-up.
- (v) Fuel cells are increasingly being used to produce energy.
Suggest two environmental disadvantages of using the conventional internal combustion engine found in cars compared to the magnesium-hydrogen peroxide fuel cell.

[7]

[Total: 20]

- 3 Pharmaceutical drugs such as aspirin and triflusal, have been found to prevent heart attacks and strokes. The active ingredients in aspirin and triflusal are the monobasic acids, **HA₁** and **HA₂**, respectively.

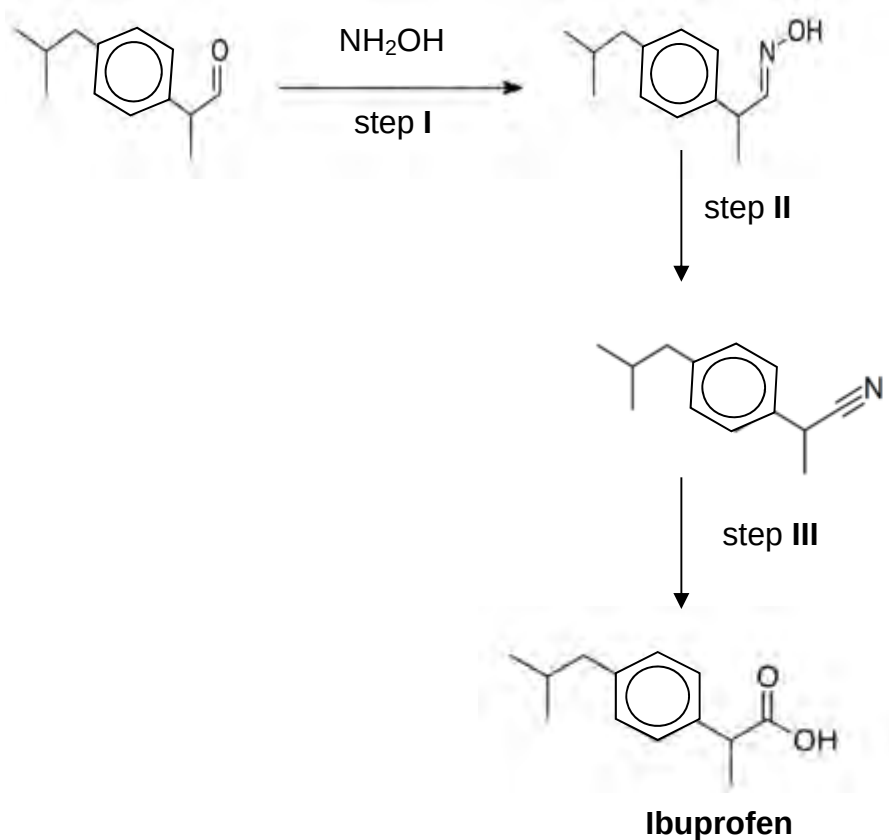
The structures and pK_a values of both acids are shown in the table below.

Compound	pK _a
 HA₁	4.80
 HA₂	3.00

- (a) (i) State and explain how the acidities of compounds **HA₁** and **HA₂** may compare with each other.
- (ii) Calculate the pH of a 0.20 mol dm⁻³ solution of compound **HA₁**. [5]
- (b) Aspirin has been found to cause stomach irritation in some people. In such cases, alternative forms of the drug have been prescribed. These alternative forms contain a *buffer* system of **HA₁** and its sodium salt.
- (i) Explain what is meant by the term, *buffer solution*.
- (ii) Calculate the pH of a buffer solution containing 20.00 cm³ of 0.20 mol dm⁻³ **HA₁** mixed with 20.00 cm³ of 0.25 mol dm⁻³ of its sodium salt.
- (iii) Calculate the pH when 5.00 cm³ of 0.10 mol dm⁻³ HCl is added to the buffer solution in **b(ii)**. [5]

- (c) Ibuprofen is a pharmaceutical drug that is manufactured for a similar purpose as aspirin, which is to relieve minor aches and pains.

Ibuprofen can be synthesised according to the following reaction scheme.

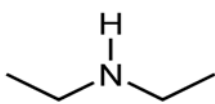


- (i) Suggest the type of reaction in step I and step II.
- (ii) Suggest the reagents and conditions for step III.
- (iii) Ibuprofen is sparingly soluble in water. Suggest an inorganic reagent that could be added to Ibuprofen to increase its solubility.

[4]

- (d) Compound **C** is used in the synthesis of a common local anesthetic drug, **D**.

When compound **C** ($\text{C}_8\text{H}_8\text{NOCl}$) is reacted with hot aqueous sodium hydroxide, compound **E** ($\text{C}_6\text{H}_7\text{N}$) and a sodium salt of acid **F** are formed. **E** reacts with chlorine water to give a tri-substituted organic product.

C reacts with alcoholic  under pressure, in the last stage of synthesis of the drug, to give **D**.

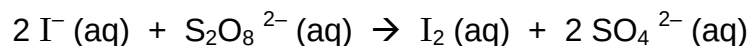
Deduce the structures of **C** to **F**, explaining the reactions that occur.

[6]

[Total: 20]

4 Transition elements such as iron and cobalt and their compounds have an important role in industry as catalysts.

(a) The reaction between iodide and peroxodisulfate ions is an example of a reaction that is catalysed by iron(II) ions, Fe^{2+} .



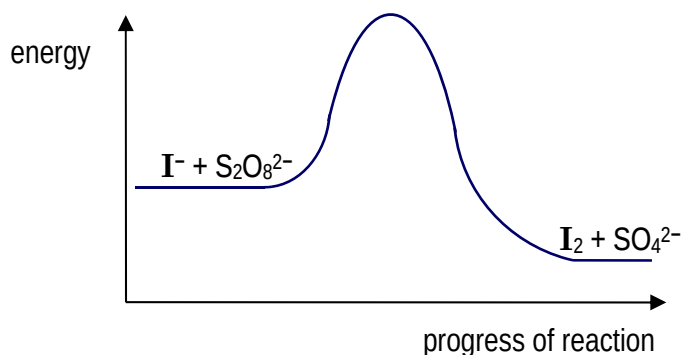
(i) The role of iron(II) ions here is described as a **homogeneous catalyst**. Explain what is meant by the term *homogeneous catalyst*.

(ii) Suggest a reason why the reaction between iodide and peroxodisulfate ions requires a catalyst.

(iii) Besides iron(II) ions, cobalt(II) ions are also suitable catalysts for the above reaction.

With reference to relevant $E^\ominus$ values in the *Data Booklet*, describe how Co^{2+} ions are able to catalyse the reaction between iodide and peroxodisulfate. Write equations where appropriate.

(iv) The reaction pathway diagram for the uncatalysed reaction is given below.



Copy the diagram and draw, on the same axes, a possible reaction pathway diagram for the catalysed reaction.

[7]

(b) The rate of a reaction can also be increased by increasing the temperature. By means of the Maxwell-Boltzmann distribution of molecular energies, explain how an increase in temperature leads to an increase in reaction rate.

[3]

- (c) Another characteristic feature of the chemistry of transition elements is their tendency to form complexes.

When aqueous cobalt(III) chloride reacts with ammonia under various conditions, compounds containing different cobalt complexes can be formed.

Two of such compounds, **X** and **Y**, have the same formula $\text{CoCl}_3(\text{NH}_3)_4$. A third compound, **Z**, has the formula $\text{CoCl}_3(\text{NH}_3)_5$.

- When treated with aqueous silver nitrate, both **X** and **Y** each form 1 mole of a white precipitate per mole of the compound whereas **Z** forms 2 moles of the white precipitate.
 - Although the cobalt complex ion in **X** and **Y** have the same formula, the complex ion in **X** does not have a net dipole moment whereas the ion in **Y** has a net dipole moment.
- (i) Identify the white precipitate that forms when **X**, **Y** and **Z** are treated with aqueous silver nitrate.
- (ii) Suggest the formulae of the cobalt complex ions in **X** and in **Z**.
- (iii) Suggest a structure for each of the cobalt complex ion in **X** and in **Y** which may be used to explain the presence or absence of a net dipole moment.

[5]

- (d) Biological catalysts such as enzymes also play an important role in industries such as food production. An example is the enzyme, *Chymosin*, which is used in the manufacture of cheese.

Chymosin is a protein molecule comprising more than 300 amino acid residues. Some of the amino acids found in *Chymosin* are shown below.

amino acid	formula of side chain (R in $\text{RCH}(\text{NH}_2)\text{CO}_2\text{H}$)
aspartic acid	$-\text{CH}_2\text{COOH}$
leucine	$-\text{CH}_2\text{CH}(\text{CH}_3)_2$
serine	$-\text{CH}_2\text{OH}$
threonine	$-\text{CH}(\text{OH})\text{CH}_3$
valine	$-\text{CH}(\text{CH}_3)_2$
lysine	$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$

- (i) With reference to relevant amino acids from the table on page 10, describe **two** forms of interaction that might be responsible for maintaining the tertiary structure of *Chymosin*. Illustrate your answer with suitable pairs of amino acids from the table on page 10.
- (ii) It has been found that *Chymosin* loses its activity under conditions of extreme pH.
With reference to the amino acid residues present in *Chymosin*, explain how conditions of extreme **high** pH leads to this loss of activity.

[5]

[Total: 20]

5 Chlorine is an important element for the chemical industry. In nature, chlorine is seldom found in its elemental form, but mostly as chlorides and chlorine oxides.

- (a) Describe and explain what you would see when solid sodium chloride and solid phosphorus pentachloride are separately added to an aqueous solution containing a few drops of Universal Indicator.

Write an equation for any reaction that occurs.

[3]

- (b) Sulfur dichloride, SCl_2 , is formed from the chlorination of elemental sulfur. It is a cherry-red liquid at room temperature and pressure, and has a boiling point of 59°C .

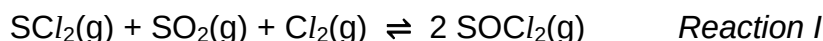
(i) Draw a dot-and-cross diagram showing the valence electrons in SCl_2 , and use the VSEPR Theory to predict the shape of SCl_2 .

(ii) Assuming ideal gas behaviour, determine the volume occupied by a sample of 1.50 g of SCl_2 at 80°C and 1 atm.

(iii) The actual volume occupied by the 1.50 g sample of SCl_2 is smaller than that determined in **b(ii)**. Suggest an explanation for this observation.

[5]

- (c) Sulfur dichloride can further react with sulfur dioxide and chlorine to form thionyl chloride, SOCl_2 , according to the equation:



When an equimolar mixture of SO_2 , Cl_2 , and SCl_2 at an initial total pressure of 108 atm was allowed to reach equilibrium at 80°C , the partial pressure of SOCl_2 is found to be 55.2 atm.

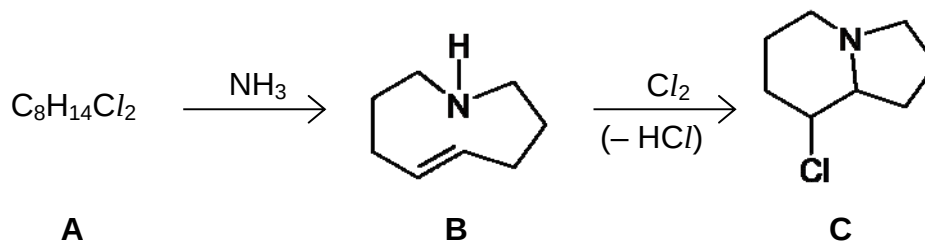
(i) Write an expression for K_p for *Reaction I*, stating its units.

(ii) Determine the equilibrium partial pressure of SO_2 , Cl_2 , and SCl_2 .

(iii) Hence, determine a value for the equilibrium constant, K_p of the reaction.

[5]

- (d) Chlorine is used extensively in organic addition and substitution reactions. The following reaction scheme shows how an aliphatic compound **A**, $C_8H_{14}Cl_2$, can be converted to the monocyclic compound **B**, and subsequently to the bicyclic product **C**.



- (i) Compound **A** exhibits geometric isomerism. Draw and label clearly the isomers of compound **A**.
- (ii) State and explain the difference in basic strength between ammonia and compound **B**.
- (iii) The reaction of compound **B** with chlorine gas begins with the electrophilic attack of chlorine on the alkene functional group.

Describe the two-step mechanism for formation of the bicyclic product **C** from compound **B**.

[7]

[Total: 20]

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